

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 8: Introduction to Trigonometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Trigonometric ratios

1. If $\sin \theta = \frac{a}{b}$, write $\cos \theta$ in terms of a and b . (a) $\frac{b}{\sqrt{b^2 - a^2}}$ (b) $\frac{b}{a}$ (c) $\frac{\sqrt{b^2 - a^2}}{b}$ (d) $\frac{a}{\sqrt{b^2 - a^2}}$

CBSE 2026, 1 mark

SOLUTION

Take a right triangle with the side opposite θ equal to a and the hypotenuse equal to b .

The adjacent side is $\sqrt{b^2 - a^2}$, so $\cos \theta = \frac{\sqrt{b^2 - a^2}}{b}$.

Answer: (c)

2. If $\tan \theta = \frac{24}{7}$, find $\sin \theta + \cos \theta$. CBSE 2026, 2 marks

SOLUTION

Take the opposite side as 24 and the adjacent side as 7. Hypotenuse = $\sqrt{24^2 + 7^2} = \sqrt{625} = 25$.

$$\sin \theta = \frac{24}{25} \text{ and } \cos \theta = \frac{7}{25}$$

Answer: $\frac{31}{25}$

3. If $\cot \theta = \frac{7}{8}$, evaluate $\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$. CBSE 2026, 2 marks

SOLUTION

Numerator = $1 - \sin^2 \theta = \cos^2 \theta$ and denominator = $1 - \cos^2 \theta = \sin^2 \theta$

$$\text{Value} = \frac{\cos^2 \theta}{\sin^2 \theta} = \cot^2 \theta = \left(\frac{7}{8}\right)^2$$

Answer: $\frac{49}{64}$

4. If $2 \tan A = 3$, find the value of $\frac{4 \sin A + 3 \cos A}{4 \sin A - 3 \cos A}$. (a) $\frac{7}{\sqrt{13}}$ (b) $\frac{1}{\sqrt{13}}$ (c) 3 (d) does not exist

CBSE 2023, 1 mark

SOLUTION

Divide the numerator and the denominator by $\cos A$: $\frac{4 \tan A + 3}{4 \tan A - 3}$

$\tan A = \frac{3}{2}$, so $4 \tan A = 6$ and the value is $\frac{6+3}{6-3} = 3$.

Answer: (c)

5. If $\sin \alpha = \frac{1}{\sqrt{2}}$ and $\cot \beta = \sqrt{3}$, find $\operatorname{cosec} \alpha + \operatorname{cosec} \beta$. CBSE 2023, 2 marks

SOLUTION

$$\operatorname{cosec} \alpha = \frac{1}{\sin \alpha} = \sqrt{2}$$

$\cot \beta = \sqrt{3}$ gives $\beta = 30^\circ$, so $\operatorname{cosec} \beta = 2$.

Answer: $2 + \sqrt{2}$

6. If $4 \tan \theta = 3$, evaluate $\frac{4 \sin \theta - \cos \theta + 1}{4 \sin \theta + \cos \theta - 1}$. CBSE 2018, 3 marks

SOLUTION

$\tan \theta = \frac{3}{4}$. Take the opposite side as 3 and the adjacent side as 4, so the hypotenuse is 5.

$$\sin \theta = \frac{3}{5} \text{ and } \cos \theta = \frac{4}{5}$$

$$\text{Numerator} = \frac{12}{5} - \frac{4}{5} + 1 = \frac{13}{5} \text{ and denominator} = \frac{12}{5} + \frac{4}{5} - 1 = \frac{11}{5}$$

Answer: $\frac{13}{11}$

2 Ratios of standard angles

7. If $\cos A = \frac{1}{2}$, find the value of $\sin^2 A + 2 \cos^2 A$. (a) $\frac{3}{2}$ (b) $\frac{5}{4}$ (c) -1 (d) $\frac{1}{2}$ CBSE 2026, 1 mark

SOLUTION

$$\cos^2 A = \frac{1}{4}, \text{ so } \sin^2 A = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\sin^2 A + 2 \cos^2 A = \frac{3}{4} + \frac{2}{4} = \frac{5}{4}$$

Answer: (b)

8. Find the acute angle θ for which $7 + 4 \sin \theta = 9$. CBSE 2025, 1 mark

SOLUTION

$$4 \sin \theta = 2, \text{ so } \sin \theta = \frac{1}{2}$$

Answer: $\theta = 30^\circ$

9. Given that $x \cos 60^\circ + y \cos 0^\circ + \sin 30^\circ - \cot 45^\circ = 5$, find the value of $x + 2y$. **CBSE 2025, 2 marks**

SOLUTION

Put in the values: $x \times \frac{1}{2} + y \times 1 + \frac{1}{2} - 1 = 5$

$\frac{x}{2} + y = \frac{11}{2}$. Multiply both sides by 2.

Answer: $x + 2y = 11$

10. Evaluate $\frac{\tan^2 60^\circ}{\sin^2 60^\circ + \cos^2 30^\circ}$. **CBSE 2025, 2 marks**

SOLUTION

$\tan 60^\circ = \sqrt{3}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$ and $\cos 30^\circ = \frac{\sqrt{3}}{2}$

Value = $\frac{3}{\frac{3}{4} + \frac{3}{4}} = \frac{3}{\frac{3}{2}}$

Answer: 2

11. Given $\cos(\alpha + \beta) = 0$, find $\cos\left(\frac{\alpha + \beta}{2}\right)$. (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$ (c) 0 (d) $\sqrt{2}$ **CBSE 2024, 1 mark**

SOLUTION

$\cos(\alpha + \beta) = 0 = \cos 90^\circ$, so $\alpha + \beta = 90^\circ$.

$\cos\left(\frac{\alpha + \beta}{2}\right) = \cos 45^\circ = \frac{1}{\sqrt{2}}$

Answer: (a)

12. Find the value of $2\sqrt{2} \cos 45^\circ \sin 30^\circ + 2\sqrt{3} \cos 30^\circ$. **CBSE 2024, 2 marks**

SOLUTION

$= 2\sqrt{2} \times \frac{1}{\sqrt{2}} \times \frac{1}{2} + 2\sqrt{3} \times \frac{\sqrt{3}}{2}$

$= 1 + 3$

Answer: 4

13. For $A = 60^\circ$ and $B = 30^\circ$, verify that $\sin(A + B) = \sin A \cos B + \cos A \sin B$. **CBSE 2024, 2 marks**

SOLUTION

LHS = $\sin(60^\circ + 30^\circ) = \sin 90^\circ = 1$

RHS = $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ = \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1$

LHS = RHS. Verified.

14. Evaluate $\frac{3}{4} \tan^2 30^\circ - \sec^2 45^\circ + \sin^2 60^\circ$. (a) -1 (b) $\frac{5}{6}$ (c) $-\frac{3}{2}$ (d) $\frac{1}{6}$ **CBSE 2023, 1 mark**

SOLUTION

$$\tan^2 30^\circ = \frac{1}{3}, \sec^2 45^\circ = 2 \text{ and } \sin^2 60^\circ = \frac{3}{4}$$

$$\text{Value} = \frac{3}{4} \times \frac{1}{3} - 2 + \frac{3}{4} = \frac{1}{4} - 2 + \frac{3}{4} = -1$$

Answer: (a)

3 Trigonometric identities

15. Given $x = h + a \cos \theta$ and $y = k + b \sin \theta$, prove that $\left(\frac{x-h}{a}\right)^2 + \left(\frac{y-k}{b}\right)^2 = 1$. **CBSE 2026, 3 marks**

SOLUTION

From the first relation, $x - h = a \cos \theta$, so $\frac{x-h}{a} = \cos \theta$.

From the second, $y - k = b \sin \theta$, so $\frac{y-k}{b} = \sin \theta$.

Squaring and adding: $\text{LHS} = \cos^2 \theta + \sin^2 \theta = 1 = \text{RHS}$. **Proved.**

16. Prove that $\frac{\tan A}{1 + \sec A} - \frac{\tan A}{1 - \sec A} = 2 \operatorname{cosec} A$. **CBSE 2026, 3 marks**

SOLUTION

$$\text{LHS} = \tan A \left[\frac{(1 - \sec A) - (1 + \sec A)}{(1 + \sec A)(1 - \sec A)} \right] = \frac{-2 \tan A \sec A}{1 - \sec^2 A}$$

Since $\sec^2 A - 1 = \tan^2 A$, the denominator is $-\tan^2 A$.

$$\begin{aligned} \text{LHS} &= \frac{-2 \tan A \sec A}{-\tan^2 A} = \frac{2 \sec A}{\tan A} = \frac{2}{\cos A} \times \frac{\cos A}{\sin A} = \frac{2}{\sin A} \\ &= 2 \operatorname{cosec} A = \text{RHS. Proved.} \end{aligned}$$

17. Simplify $\tan^2 \theta - \left(\frac{1}{\cos \theta} \times \sec \theta \right)$. **CBSE 2025, 1 mark**

SOLUTION

$\frac{1}{\cos \theta} = \sec \theta$, so the bracket is $\sec^2 \theta$.

$$\tan^2 \theta - \sec^2 \theta = -(\sec^2 \theta - \tan^2 \theta) = -1$$

Answer: -1

18. Prove that $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$. **CBSE 2025, 3 marks**

SOLUTION

Write everything in terms of $\sin \theta$ and $\cos \theta$.

$$\text{First term: } \frac{\sin \theta}{\cos \theta} \div \frac{\sin \theta - \cos \theta}{\sin \theta} = \frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)}$$

$$\text{Second term: } \frac{\cos \theta}{\sin \theta} \div \frac{\cos \theta - \sin \theta}{\cos \theta} = -\frac{\cos^2 \theta}{\sin \theta (\sin \theta - \cos \theta)}$$

$$\text{LHS} = \frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta \cos \theta (\sin \theta - \cos \theta)}$$

$$\begin{aligned} \text{Using } a^3 - b^3 &= (a - b)(a^2 + ab + b^2): \text{LHS} = \frac{\sin^2 \theta + \sin \theta \cos \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \frac{1 + \sin \theta \cos \theta}{\sin \theta \cos \theta} \\ &= \frac{1}{\sin \theta \cos \theta} + 1 = 1 + \sec \theta \operatorname{cosec} \theta = \text{RHS. Proved.} \end{aligned}$$

19. Prove that $\frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A} = \frac{2}{2 \sin^2 A - 1}$. **CBSE 2025, 3 marks**

SOLUTION

Take the common denominator $(\sin A - \cos A)(\sin A + \cos A) = \sin^2 A - \cos^2 A$.

$$\text{Numerator} = (\sin A + \cos A)^2 + (\sin A - \cos A)^2 = 2 \sin^2 A + 2 \cos^2 A = 2$$

$$\begin{aligned} \text{LHS} &= \frac{2}{\sin^2 A - \cos^2 A} = \frac{2}{\sin^2 A - (1 - \sin^2 A)} \\ &= \frac{2}{2 \sin^2 A - 1} = \text{RHS. Proved.} \end{aligned}$$

20. Given $\sec \theta - \tan \theta = m$, find $\sec \theta + \tan \theta$. (a) $1 - \frac{1}{m}$ (b) $m^2 - 1$ (c) $\frac{1}{m}$ (d) $-m$ **CBSE 2024, 1 mark**

SOLUTION

$$\sec^2 \theta - \tan^2 \theta = 1, \text{ so } (\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1.$$

$$\sec \theta + \tan \theta = \frac{1}{\sec \theta - \tan \theta} = \frac{1}{m}$$

Answer: (c)

21. Prove the identity $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$. **CBSE 2024, 3 marks**

SOLUTION

$$\text{Write } s = \sin \theta \text{ and } c = \cos \theta. \text{ Then } \frac{\tan \theta}{1 - \cot \theta} = \frac{s/c}{(s-c)/s} = \frac{s^2}{c(s-c)} \text{ and } \frac{\cot \theta}{1 - \tan \theta} = \frac{c/s}{(c-s)/c} = -\frac{c^2}{s(s-c)}.$$

$$\begin{aligned} \text{LHS} &= \frac{s^3 - c^3}{sc(s-c)} = \frac{(s-c)(s^2 + sc + c^2)}{sc(s-c)} = \frac{s^2 + c^2 + sc}{sc} \\ &= \frac{1 + sc}{sc} = \frac{1}{sc} + 1 \\ &= 1 + \sec \theta \operatorname{cosec} \theta = \text{RHS} \end{aligned}$$

- 22.** If $\sin \theta + \cos \theta = \sqrt{3}$, find the value of $\sin \theta \cos \theta$. **CBSE 2023, 2 marks**

SOLUTION

Square both sides: $\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = 3$

$$1 + 2 \sin \theta \cos \theta = 3$$

Answer: $\sin \theta \cos \theta = 1$

- 23.** Prove that $\frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1} = \frac{1 + \sin \theta}{\cos \theta}$. **CBSE 2023, 3 marks**

SOLUTION

In the numerator, replace 1 by $\sec^2 \theta - \tan^2 \theta$.

$$\text{Numerator} = (\sec \theta + \tan \theta) - (\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = (\sec \theta + \tan \theta)(1 - \sec \theta + \tan \theta)$$

The second bracket is the same as the denominator, so LHS = $\sec \theta + \tan \theta$.

$$= \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} = \frac{1 + \sin \theta}{\cos \theta} = \text{RHS. Proved.}$$

- 24.** Prove that $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = 7 + \tan^2 \theta + \cot^2 \theta$. **CBSE 2019, 3 marks**

SOLUTION

$$\text{LHS} = \sin^2 \theta + \operatorname{cosec}^2 \theta + 2 \sin \theta \operatorname{cosec} \theta + \cos^2 \theta + \sec^2 \theta + 2 \cos \theta \sec \theta$$

$$\sin \theta \operatorname{cosec} \theta = 1 \text{ and } \cos \theta \sec \theta = 1, \text{ so LHS} = (\sin^2 \theta + \cos^2 \theta) + 4 + \operatorname{cosec}^2 \theta + \sec^2 \theta$$

$$= 1 + 4 + (1 + \cot^2 \theta) + (1 + \tan^2 \theta)$$

$$= 7 + \tan^2 \theta + \cot^2 \theta = \text{RHS. Proved.}$$

- 25.** Prove that $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$. **CBSE 2019, 3 marks**

SOLUTION

$$\begin{aligned} \text{LHS} &= \left(\frac{\sin A + \cos A - 1}{\sin A} \right) \left(\frac{\cos A + \sin A + 1}{\cos A} \right) \\ &= \frac{(\sin A + \cos A)^2 - 1}{\sin A \cos A} = \frac{\sin^2 A + \cos^2 A + 2 \sin A \cos A - 1}{\sin A \cos A} \\ &= \frac{2 \sin A \cos A}{\sin A \cos A} = 2 = \text{RHS. Proved.} \end{aligned}$$

- 26.** Prove that $\frac{\sin A - \cos A + 1}{\sin A + \cos A - 1} = \frac{1}{\sec A - \tan A}$. **CBSE 2019, 4 marks**

SOLUTION

$$\text{Divide the numerator and the denominator by } \cos A: \text{LHS} = \frac{\tan A - 1 + \sec A}{\tan A + 1 - \sec A}$$

$$\text{In the denominator, write } 1 = \sec^2 A - \tan^2 A = (\sec A - \tan A)(\sec A + \tan A).$$

$$\text{Denominator} = (\tan A - \sec A) + (\sec A - \tan A)(\sec A + \tan A) = (\sec A - \tan A)(\sec A + \tan A - 1)$$

$$\text{LHS} = \frac{\tan A + \sec A - 1}{(\sec A - \tan A)(\sec A + \tan A - 1)} = \frac{1}{\sec A - \tan A} = \text{RHS. Proved.}$$

27. Prove that $\frac{\sin A - 2 \sin^3 A}{2 \cos^3 A - \cos A} = \tan A$. **CBSE 2018, 4 marks**

SOLUTION

$$\text{LHS} = \frac{\sin A(1 - 2 \sin^2 A)}{\cos A(2 \cos^2 A - 1)}$$

Using $1 = \sin^2 A + \cos^2 A$: $1 - 2 \sin^2 A = \cos^2 A - \sin^2 A$ and $2 \cos^2 A - 1 = \cos^2 A - \sin^2 A$.

The two brackets are equal and cancel: $\text{LHS} = \frac{\sin A}{\cos A} = \tan A = \text{RHS}$. **Proved.**

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